Universal Origins: The Zero-Sum Constraint Matrix Formulation of Cosmogenesis

Infinite-Dimensional Linear Algebra Structure of the Pre-Planck Vacuum, Its QGP Free-Streaming Regime, and the Flat-Space Equation of Motion

Romeo Matshaba

Department of Physics, University of South Africa, Pretoria, South Africa

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Abstract

We develop a matrix-algebraic framework for cosmogenesis based on a single structural constraint on the universe's density operator. Let $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ be the Fock space of possible universe states, $\hat{\rho}(t) \in \mathcal{B}(\mathcal{F})$ the global density operator, and $\{\hat{Q}_k\}$ the complete set of Standard Model conserved charges. We impose one axiom: $\mathrm{Tr}[\hat{\rho}(t)\hat{Q}_k] = 0$ for all k and all times t — a constraint that every element of the observable charge vector equals zero, $\mathbf{Q}[\hat{\rho}] = \mathbf{0}_\infty$.

Working strictly within this axiom, the Heisenberg uncertainty principle, and the Standard Model degree-of-freedom count $g_* = 106.75$, we derive without additional assumptions: the radiation equation of state $w = 1/3$; the Hubble rate $H(t_{\mathrm{Pl}}) = 1/(2t_{\mathrm{Pl}})$; exact spatial flatness $\Omega = 1$; all five hot big-bang initial conditions; and the Planck-cell energy range $E_{\mathrm{cell}} \in [35.86 \pm 0.61,\ 53.375]\, E_{\mathrm{Pl}}$ set by two independent Heisenberg bounds. The lower bound uses the shortest QCD timescale $\tau_{gg} = 1.489\, t_{\mathrm{Pl}}$, computed from running $\alpha_s(E_{\mathrm{Pl}}/2) = 0.01055$ (PDG 2024). A central new result is that the emergent QGP is in the *free-streaming* regime with mean free path $\lambda_{\mathrm{mfp}} \approx 85$–$94\, \ell_{\mathrm{Pl}} \gg \ell_{\mathrm{Pl}}$; thermalisation requires $\tau_{\mathrm{th}} \approx 85$–$94\, t_{\mathrm{Pl}}$, not instantaneous. The TUO–FRW junction occurs at $t_{\mathrm{junc}} \in [17, 94]\, t_{\mathrm{Pl}}$. The flat-space equation of motion $\ddot{\sigma} = c^2/\sigma$ is derived from Newton's second law with SR inertia (no GR); its deceleration parameter $q(t) = -(\ell_{\mathrm{Pl}}/ct)^2$ satisfies $q(t_{\mathrm{Pl}}) = -1$ exactly (de Sitter equivalent). The pre-emergence temperature is $T_{\mathrm{TUO}} = (15/\pi^2)^{1/4} T_{\mathrm{Pl}} \approx 1.108\, T_{\mathrm{Pl}}$, g_*-independent; the factor $15/\pi^2$ is an energy-density ratio, not a particle-number ratio. Six open problems are precisely stated.

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1 Introduction

The two deepest unanswered questions in cosmology are: *why did the universe begin?* and *why did it begin with those particular initial conditions?* The second question is, in a sense, more tractable. Standard Big Bang cosmology describes the evolution of the universe extraordinarily well from $t \sim t_{\mathrm{Pl}}$ onwards, but its initial conditions — the equation of state $w = 1/3$, the spatial flatness $\Omega = 1$, and the temperature $T \approx T_{\mathrm{Pl}}$ — are inputs rather than outputs [7, 8]. The inflationary scenario addresses some of these by introducing an additional scalar field, but it does not eliminate the question of pre-inflationary initial conditions [4, 5].

The present paper develops the Theory of Universal Origins (TUO), which proposes that the relevant initial conditions follow mathematically from a single structural constraint on the universe's density operator in an infinite-dimensional Fock space. The constraint is: the trace of the density operator against any conserved charge operator must equal zero at all times. Symbolically, the observable charge vector equals the zero object — the infinite zero matrix. Everything, in a precise operator-algebraic sense, sums to nothing.

The mathematical language is infinite-dimensional linear algebra. The density operator is an element of $\mathcal{B}(\mathcal{F})$; the charge operators are its diagonal entries in an appropriate basis; the constraint is a collection of linear equations on those entries. This language makes the structure of the constraint — and what it does and does not determine — maximally transparent.

What this paper does not claim.

- We do not claim to explain the number of fermion generations.
- We do not derive the baryon asymmetry η quantitatively.
- We do not explain the CMB power spectrum.
- We do not replace general relativity; a companion framework (Quantum Gravity Dynamics) recovering GR from the flat background is a separate project.
- We do not derive dark matter or dark energy from these axioms.
- Where a result requires physics beyond the two axioms, we explicitly label it as a *possible mechanism* or *open problem*.

2 Mathematical Foundations: Fock Space and the Zero Matrix

2.1 The zero object

The central object of TUO is the infinite zero matrix:

$$\mathbf{0}_\infty = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \in \mathcal{B}(\mathcal{F}). \tag{1}$$

The claim of TUO is that the observable content of the universe, when correctly assembled, equals $\mathbf{0}_\infty$ at every instant. This is not a dynamical statement; it is an *existence filter*.

2.2 Fock space structure

Definition 2.1 (Universe Fock space)**.** Let $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ be the bosonic-fermionic Fock space over the single-particle Hilbert space $\mathcal{H}_1$ of Standard Model fields. States $|\Psi\rangle \in \mathcal{F}$ are superpositions of n-particle sectors:

$$|\Psi\rangle = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \end{pmatrix}, \quad \sum_{n=0}^{\infty} |c_n|^2 = 1. \tag{2}$$

2.3 The density operator

The universe is described by a density operator $\hat{\rho}(t) \in \mathcal{B}(\mathcal{F})$ satisfying

$$\hat{\rho} = \hat{\rho}^\dagger, \quad \mathrm{Tr}[\hat{\rho}] = 1, \quad \langle\psi|\hat{\rho}|\psi\rangle \geq 0 \ \ \forall\, |\psi\rangle. \tag{3}$$

In a basis $\{|n\rangle\}_{n=0}^{\infty}$ of $\mathcal{F}$:

$$\hat{\rho} = \sum_{m,n=0}^{\infty} \rho_{mn} |m\rangle\langle n| = \begin{pmatrix} \rho_{00} & \rho_{01} & \rho_{02} & \cdots \\ \rho_{10} & \rho_{11} & \rho_{12} & \cdots \\ \rho_{20} & \rho_{21} & \rho_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{4}$$

2.4 Charge operators and the trace form

Each conserved charge corresponds to a Hermitian operator $\hat{Q}_k \in \mathcal{B}(\mathcal{F})$ with eigenvalues $\{q_{kn}\}_{n\geq 0}$. In its eigenbasis:

$$\hat{Q}_k = \mathrm{diag}(q_{k0},\, q_{k1},\, q_{k2},\, \ldots). \tag{5}$$

The expectation value is the trace:

$$\langle \hat{Q}_k \rangle = \mathrm{Tr}[\hat{\rho}\, \hat{Q}_k] = \sum_{n=0}^{\infty} \rho_{nn}\, q_{kn} = \mathbf{p} \cdot \mathbf{q}_k, \tag{6}$$

an *infinite dot product* of the occupation-probability vector $\mathbf{p} = (\rho_{00}, \rho_{11}, \ldots)$ against the eigenvalue vector $\mathbf{q}_k$.

3 The Two Axioms

Axiom 1 (Flat Background)**.** The pre-emergence arena is (3+1)-dimensional Minkowski spacetime: $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. No curvature, no preferred time, no pre-existing matter.

Remark 3.1. Axiom 1 is foundational and self-standing. It is not derived from general relativity or any other framework. Why four dimensions? Why Minkowski? These questions are not answered here; the axiom names the background and treats that choice as primitive.

Axiom 2 (Global Zero-Sum Constraint)**.** For all conserved charges $\hat{Q}_k$ and all times t:

$$\boxed{\mathrm{Tr}\big[\hat{\rho}(t)\, \hat{Q}_k\big] = 0 \quad \forall\, k,\ \forall\, t.} \tag{7}$$

Equivalently, the observable charge vector

$$\mathbf{Q}[\hat{\rho}] = \big(\mathrm{Tr}[\hat{\rho}\hat{Q}_E],\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{P^\mu}],\ \mathrm{Tr}[\hat{\rho}\hat{Q}_J],\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{\mathrm{em}}],\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{B-L}],\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{\mathbf{C}}]\big)^\top = \mathbf{0}_{12\times 1}. \tag{8}$$

Remark 3.2. Energy (the Hamiltonian) is one conserved charge. Axiom 2 is therefore strictly stronger than requiring zero total energy alone: it also sets net momentum, angular momentum, and every gauge charge to zero. The pre-emergence state is a quantum nothing in every sense that Noether's theorem can assign a conserved number to. Imposing only zero energy would leave, for example, a possible nonzero angular momentum — which would carry a preferred axis into the emerged universe. The full zero-sum axiom eliminates all such remnants simultaneously.

Remark 3.3. Equation (7) is an infinite linear constraint on $\hat{\rho}$: substituting (6), it reads $\sum_{n\geq 0} \rho_{nn} q_{kn} = 0$ for each k. This defines a hyperplane $\mathcal{Z}_k \subset \mathcal{B}(\mathcal{F})$, and the set of physically allowed density operators is $\mathcal{Z} = \bigcap_k \mathcal{Z}_k \cap \{\text{density operators}\}$.

Remark 3.4. Axiom 1 and Axiom 2 are the *only* postulates of TUO. Every other result in this paper is derived from these two axioms plus Standard Model particle content. Where additional physical input is invoked, it is explicitly indicated.

4 Vacuum State and the Zero-Sum Filter

4.1 The vacuum

Definition 4.1 (Vacuum state)**.** The ground state prior to any excitation:

$$|0\rangle = (1, 0, 0, \ldots)^\top, \qquad \hat{\rho}_0 = |0\rangle\langle 0| = \mathrm{diag}(1, 0, 0, \ldots). \tag{9}$$

Lemma 4.2. *The vacuum satisfies Axiom 2:* $\mathrm{Tr}[\hat{\rho}_0 \hat{Q}_k] = q_{k0} = 0$.

4.2 Single-particle prohibition

Theorem 4.3. *A pure single-particle state* $|1; q\rangle$ *with charge eigenvalue* $q \neq 0$ *violates Axiom 2.*

Proof. $\mathrm{Tr}[\hat{\rho}_1 \hat{Q}] = q \neq 0$ for $\hat{\rho}_1 = \mathrm{diag}(0, 1, 0, \ldots)$, $\hat{Q} = \mathrm{diag}(0, q, 0, \ldots)$. □

4.3 Pair configurations

Theorem 4.4. *A superposition of states with opposite charge eigenvalues satisfies Axiom 2 for that charge.*

Proof. Let $|\psi\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ with charges $\pm q$. Then $\mathrm{Tr}[\hat{\rho}_{\mathrm{pair}} \hat{Q}] = \frac{1}{2}(q - q) = 0$. □

Remark 4.5. The zero-sum constraint acts as a *linear filter*: it admits any state in the kernel of the charge functional. This is the structural reason that matter-antimatter pairs are the generic fluctuation, while matter-only configurations are special.

5 The Complete Charge Equation

5.1 Hamiltonian decomposition and energy balance

The total Hamiltonian splits as $\hat{H} = \hat{H}_{\mathrm{matter}} + \hat{H}_{\mathrm{field}}$.

Theorem 5.1 (Energy balance)**.** *Axiom 2 applied to $\hat{H}$ requires $E_{\text{matter}} + E_{\text{field}} = 0$ exactly.*

Proof. $0 = \mathrm{Tr}[\hat{\rho}\hat{H}] = \mathrm{Tr}[\hat{\rho}\hat{H}_{\mathrm{m}}] + \mathrm{Tr}[\hat{\rho}\hat{H}_{\mathrm{f}}] = E_{\mathrm{m}} + E_{\mathrm{f}}$. □

5.2 The complete zero equation

All 12 conserved-charge observables satisfy

$$\underbrace{\begin{pmatrix} \mathrm{Tr}[\hat{\rho}\hat{Q}_E] \\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{P^\mu}] \\ \mathrm{Tr}[\hat{\rho}\hat{Q}_J] \\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{\mathrm{em}}] \\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{B-L}] \\ \mathrm{Tr}[\hat{\rho}\hat{Q}_{\mathbf{C}}] \end{pmatrix}}_{\mathbf{Q}[\hat{\rho}] \in \mathbb{R}^{12}} = \mathbf{0}_{12\times 1}. \tag{10}$$

This is the *central equation of TUO*: the universe's full observable content equals the infinite zero. The field contribution is whatever it must be to maintain the balance; there is no freedom to adjust it independently.

5.3 Time evolution preserves zero-sum

Theorem 5.2. *Unitary evolution preserves Axiom 2 if and only if $[\hat{H}, \hat{Q}_k] = 0$ for all k, i.e., each charge is conserved.*

Proof. $\frac{d}{dt}\mathrm{Tr}[\hat{\rho}\hat{Q}_k] = -\frac{i}{\hbar}\mathrm{Tr}[\hat{\rho}[\hat{Q}_k, \hat{H}]] = 0 \iff [\hat{H}, \hat{Q}_k] = 0.$ □

5.4 Equalities in fundamental equations as a consequence

A striking feature of fundamental physics is that governing equations are *equalities*: $G_{\mu\nu} = \kappa T_{\mu\nu}$, not inequalities. TUO provides a structural explanation.

Axiom 2 states $\mathrm{Tr}[\hat{\rho}\hat{Q}_k] = 0$ *exactly.* For any local charge density $\mathcal{Q}_k(x)$, $\int \mathcal{Q}_k(x)\, d^3x = 0$ at all times. Localising this balance pointwise requires

$$\mathcal{Q}_k^{\text{matter}}(x) + \mathcal{Q}_k^{\text{field}}(x) = 0 \quad \forall\, x. \tag{11}$$

The = sign is not empirical; it is forced. Three consequences:

Einstein field equations. Theorem 5.1 gives $E_{\mathrm{m}} + E_{\text{grav}} = 0$ globally. Localising and applying the Lovelock theorem [9] (unique rank-2 symmetric divergence-free tensor from the metric):

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}. \tag{12}$$

The equality sign is the local expression of Axiom 2.

Maxwell's equations. For the EM charge: $\partial_\mu F^{\mu\nu} = J^\nu/\varepsilon_0$. An inequality would mean the field carries more charge than the source, violating Axiom 2.

Schrödinger equation. Conservation $\frac{d}{dt}\int |\psi|^2 d^3x = 0$ requires $i\hbar\,\partial_t\psi = \hat{H}\psi$ as the unique linear first-order evolution that achieves it.

6 The Selection Problem: Why Generic Fluctuations Cannot Become a Universe

Axiom 2 permits a vast class of zero-sum fluctuations. Theorems 4.3–4.4 show that any superposition whose charges cancel is admitted. The vacuum fluctuates. The question is which fluctuations can survive long enough and carry enough energy to become a universe. Two explicit barriers prevent generic fluctuations from doing so.

6.1 Barrier I: The energy gap

A typical virtual pair carries energy $E_{e^+e^-} = 2m_ec^2 \approx 1.64 \times 10^{-13}$ J. The energy required to seed a Planck-scale cell capable of expanding into a universe is (Section 12):

$$E_{\text{cell}} = g_* \cdot \frac{E_{\text{Pl}}}{2} \approx 1.04 \times 10^{11}\ \text{J}. \tag{13}$$

The ratio is

$$\frac{E_{e^+e^-}}{E_{\text{cell}}} \approx 1.6 \times 10^{-24}. \tag{14}$$

A generic virtual pair carries 24 orders of magnitude too little energy. No accumulation mechanism exists within a single Planck-time window.

6.2 Barrier II: The annihilation timescale

At Planck energies, $\Delta t \sim t_{\text{Pl}} \approx 5.4 \times 10^{-44}$ s. Within this window, a generic pair appears and annihilates before any structure can form: $e^+e^- \to \gamma\gamma$, $q\bar{q} \to gg$, etc. Every antiparticle present opens an annihilation channel.

6.3 The resolution: the maximum fluctuation

Theorem 6.1 (Maximum fluctuation uniquely overcomes both barriers)**.** *The configuration of all SM matter — quarks and leptons, three generations, all colours and flavours — simultaneously co-located at one Planck cell with no antiparticles:*

(i) satisfies Axiom 2;

(ii) carries energy $E_{\text{cell}} = g_* \cdot E_{\text{Pl}}/2$*;*

(iii) has no annihilation channel and is kinematically stable.

Proof. (i) $B - L = 0$ and $Q = 0$ per generation (Theorem 8.1); $E_{\text{grav}} = -E_{\text{cell}}$ (Theorem 5.1). (ii) Each of g_* modes carries $E_{\text{Pl}}/2$. (iii) Zero antiparticle occupation $n_{\bar{f}} = 0$; no SM process produces $\bar{f}$ from $n_{\bar{f}} = 0$ without violating B or L conservation. □

Remark 6.2. The resolution is kinematic, not a timescale argument. Annihilation requires an antiparticle. The maximum fluctuation contains none. This is what the No-Annihilation Theorem (Theorem 8.3) makes precise.

Remark 6.3. This is the creative core of TUO. The zero-sum vacuum fluctuates continuously; generically those fluctuations are too small and too short-lived. The unique exception is the configuration that carries all SM content, simultaneously, co-located: it is energetically sufficient and kinematically stable. It is the only zero-sum fluctuation the vacuum can produce that is capable of becoming a universe. The number $g_* = 106.75$ enters not as a free parameter but as the total count of SM degrees of freedom that must emerge together for either barrier to be overcome.

7 Configuration Space and Frozen Constants

7.1 The space of stable configurations

A configuration is a choice of field content (determining g_*), charge assignments (satisfying Axiom 2 with no antiparticles), and a mass spectrum. The zero-sum filter admits a matter-only configuration if and only if it satisfies $Q = 0$ and $B - L = 0$ per generation-like block. These are the SU(5) anomaly-cancellation conditions, which the SM satisfies exactly.

Other configurations with $N_{\text{gen}} \neq 3$ also satisfy them (the identity holds per generation independently).

Table 1: Stable zero-sum configurations. Each row satisfies $Q = 0$, $B - L = 0$, and the no-annihilation condition. N_ν measured at LEP [3]: only $N_{\text{gen}} = 3$ is consistent. Note: g_* here counts fermion generations only; the SM boson sector adds $N_B = 28$, giving the full $g_* = 106.75$ for $N_{\text{gen}} = 3$.

N_{gen}	g_* (fermions)	$E_{\text{cell}}/E_{\text{Pl}}$	E_{cell} (J)
1	42.00	21.00	4.11×10^{10}
2	56.00	28.00	5.48×10^{10}
3	**70.00**	**35.00**	$\mathbf{6.85 \times 10^{10}}$
4	84.00	42.00	8.22×10^{10}
5	98.00	49.00	9.58×10^{10}

Remark 7.1. Each row is a logically consistent possible universe. TUO does not derive $N_{\text{gen}} = 3$; this is Open Problem 23.5. The LEP measurement $N_\nu = 2.984 \pm 0.008$ selects $N_{\text{gen}} = 3$ from this table observationally.

7.2 What freezes at emergence

When a configuration emerges, three classes of quantities are fixed.

Class I — Axiom-fixed. The speed c is fixed by Axiom 1 (Minkowski light-cone structure). $\hbar$ is required by the Heisenberg bound in Axiom 2. Newton's constant G is determined by the identity $GM_{\text{Pl}}^2 = \hbar c$ forced by Theorem 5.1: these are consequences of the axioms, not free parameters.

Class II — Configuration-fixed dimensionless ratios. $\alpha \approx 1/137$, α_s, and mass ratios m_f/M_{Pl} are fixed when a configuration is selected. A universe with $N_{\text{gen}} = 4$ would generically have different values.

Class III — Emergent effective constants. Λ_{QCD}, the Higgs vev, and Λ_{cosm} are dynamically determined within a configuration from Class II.

Remark 7.2. A common intuition is that c could be different in another universe. TUO refines this: c as a speed limit cannot vary because it is fixed by Axiom 1. What can vary is the ratio $m_e c^2/E_{\text{Pl}}$ (the electron is lighter or heavier relative to the Planck scale), which would alter chemistry, not spacetime geometry.

8 Standard Model Content and the $B - L$ Constraint

Theorem 8.1 ($B - L = 0$ and $Q = 0$ per generation)**.** *The SM three-generation fermion content satisfies $Q = 0$ and $B - L = 0$ per generation as an algebraic identity.*

Proof. In generation g: $B_g = 2 \times 3 \times \frac{1}{3} = 2$; $L_g = 2$; $B_g - L_g = 0$. $Q_g = 3(\frac{2}{3}) + 3(-\frac{1}{3}) + (-1) + 0 = 2 - 1 - 1 = 0$. Identical for $g = 1, 2, 3$. □

Remark 8.2. This is the SU(5) anomaly-cancellation condition. TUO does not derive the SM content; rather, the SM satisfies TUO's constraint. Whether TUO can derive the SM from first principles is an open question (Section 23).

Theorem 8.3 (No-Annihilation)**.** *A configuration satisfying $Q = 0$, $B - L = 0$, and containing no antiparticles has no SM annihilation channel.*

Proof. Every SM annihilation requires a particle-antiparticle pair. With $n_{\bar{f}} = 0$ for all $\bar{f}$, no such pair exists. No SM process produces $\bar{f}$ without violating B or L. The argument is kinematic; it makes no claim about timescales. □

9 Volume Convention and Self-Consistency

Every quantitative result depends on the definition of the Planck volume. Two natural choices exist: the sphere $V_{\rm sph} = (4\pi/3)\ell_{\rm Pl}^3$ and the cube $V_{\rm Pl} = \ell_{\rm Pl}^3$.

The cube is the correct choice for two independent reasons. First, the $15/\pi^2$ identity of Theorem 13.1 requires

$$(k_{\rm B}T_{\rm TUO})^4 = \frac{15}{\pi^2}\,(k_{\rm B}T_{\rm Pl})^4, \tag{15}$$

which holds if and only if $V_{\rm Pl} = \ell_{\rm Pl}^3$; the sphere gives the different factor $90/(8\pi^3) \approx 0.363$. Second, $\ell_{\rm Pl}^3$ is the natural mode-counting volume in a 3D Fourier basis.

Quantity	Cube ($\ell_{\rm Pl}^3$)	Sphere ($\frac{4\pi}{3}\ell_{\rm Pl}^3$)	Qualitative result
$\tau_{gg}/t_{\rm Pl}$	1.489	6.24	free-streaming ✓
$T_{\rm lower}/T_{\rm Pl}$	1.005	0.489	near $T_{\rm Pl}$ ✓
$T_{\rm upper}/T_{\rm Pl}$	1.108	0.776	$= T_{\rm TUO}$ ✓
$\lambda_{\rm mfp}/\ell_{\rm Pl}$	85–94	∼290–450	$\gg \ell_{\rm Pl}$ ✓
$t_{\rm junc}/t_{\rm Pl}$	[17, 94]	[17, 290]	$> t_{\rm Pl}$ ✓
Factor	$15/\pi^2$	$90/(8\pi^3)$	

Both choices give free-streaming, $q(t_{\rm Pl}) = -1$, and all qualitative results. The cube is used throughout.

10 Zero Total Energy

Theorem 10.1 (Zero Total Energy)**.** $E_{\rm total} = 0$ *exactly.*

Proof. The Heisenberg minimum energy is $E_{\rm matter} = E_{\rm Pl}/2$. The gravitational binding energy is $E_{\rm grav} = -GM_{\rm Pl}^2/(2\ell_{\rm Pl})$. By the definitional identity $GM_{\rm Pl}^2 = \hbar c$ (consequence of Theorem 5.1): $E_{\rm grav} = -\hbar c/(2\ell_{\rm Pl}) = -E_{\rm Pl}/2$. Sum: $E_{\rm total} = E_{\rm Pl}/2 - E_{\rm Pl}/2 = 0$. □

This is an algebraic identity. Numerically: $GM_{\rm Pl}^2/(\hbar c) = 1.000\,000\,000\,000\,000$ to 15 significant figures (machine double precision). The universe contains 1.044×10^{11} J of matter energy per Planck cell at emergence, and an equal and opposite gravitational energy. The net is exactly zero: the apparent richness of the universe is a rearrangement of nothing.

11 Energy from Heisenberg Uncertainty: The Emergence Configuration

11.1 The energy matrix at $t = t_{\mathrm{Pl}}$

The Heisenberg time-energy uncertainty with $\Delta t = t_{\mathrm{Pl}}$:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2} \implies \Delta E_{\min} = \frac{\hbar}{2t_{\mathrm{Pl}}} = \frac{E_{\mathrm{Pl}}}{2}. \tag{16}$$

At $T = T_{\mathrm{Pl}}$, all SM fields are relativistic. The effective relativistic degrees of freedom are

$$g_* = N_B + \tfrac{7}{8} N_F = 28 + \tfrac{7}{8}(90) = 106.75, \tag{17}$$

where $N_B = 28$ (gauge and Higgs bosons) and $N_F = 90$ (fermionic polarisations and colour states).

The Hamiltonian for the emergence configuration restricted to g_* modes at their Heisenberg minimum energies:

$$\hat{H}_{\mathrm{emergence}} = \mathrm{diag}\underbrace{\left(0,\ \tfrac{E_{\mathrm{Pl}}}{2},\ \tfrac{E_{\mathrm{Pl}}}{2},\ \ldots,\ \tfrac{E_{\mathrm{Pl}}}{2}\right)}_{g_* \text{ copies of } E_{\mathrm{Pl}}/2} \oplus \mathbf{0}_{\infty - g_*}. \tag{18}$$

Theorem 11.1 (Energy per Planck cell)**.**

$$\boxed{E_{\mathrm{cell}} = g_* \cdot \frac{E_{\mathrm{Pl}}}{2} = 53.375\, E_{\mathrm{Pl}} \approx 1.044 \times 10^{11}\ \mathrm{J}.} \tag{19}$$

$E_{\mathrm{grav}} = -E_{\mathrm{cell}}$ *exactly (Theorem 5.1).*

11.2 Comparison with the thermal plasma

The energy density of a thermal plasma at $T = T_{\mathrm{Pl}}$ in volume ℓ_{Pl}^3:

$$E_{\mathrm{std}} = \frac{\pi^2}{30}\, g_* \, \frac{(k_{\mathrm{B}} T_{\mathrm{Pl}})^4}{(\hbar c)^3}\, \ell_{\mathrm{Pl}}^3 = 6.87 \times 10^{10}\ \mathrm{J}. \tag{20}$$

Their ratio:

$$\frac{E_{\mathrm{cell}}}{E_{\mathrm{std}}} = \frac{15}{\pi^2} \approx 1.52. \tag{21}$$

This factor is a pure dimensionless number independent of all SM parameters and of g_* (which cancels exactly). Its sole origin is the ratio of the Heisenberg minimum energy to the Stefan-Boltzmann radiation constant.

12 Energy Range from Two Heisenberg Bounds

Theorem 12.1 (Energy Range)**.** *The Planck-cell energy satisfies*

$$E_{\text{cell}} \in [35.86 \pm 0.61,\ 53.375]\, E_{\text{Pl}}. \tag{22}$$

12.1 Upper bound (Heisenberg position)

Localising all g_* modes within one Planck length gives $\Delta p_{\min} = \hbar/(2\ell_{\text{Pl}})$. The corresponding energy per mode is $E_{\text{Pl}}/2$, yielding the Theorem 11.1 value $E_{\text{cell}}^{\text{upper}} = 53.375\, E_{\text{Pl}}$ (exact).

12.2 Lower bound from QCD interaction time

The fastest QCD process in the cell is gluon–gluon scattering with cross section

$$\sigma_{gg} = \frac{9\pi\alpha_s^2}{s}, \qquad s = \big(E_{\text{Pl}}/\hbar c\big)^2. \tag{23}$$

The 1-loop running coupling at $\mu = E_{\text{Pl}}/2$ (PDG 2024):

$$\alpha_s\big(\tfrac{E_{\text{Pl}}}{2}\big) = \frac{\alpha_s(M_Z^2)}{1 + \frac{b_0}{2\pi}\alpha_s(M_Z^2)\ln\big(\mu^2/M_Z^2\big)} = 0.01055 \pm 0.0002, \tag{24}$$

with $b_0 = 11 - 2n_f/3 = 7$, $\alpha_s(M_Z) = 0.1179$, $M_Z = 91.1876\,\text{GeV}$. Using Planck density $n = g_*/\ell_{\text{Pl}}^3$ and relative velocity $v_{\text{rel}} = 2c$:

$$\tau_{gg} = \frac{1}{n\,\sigma_{gg}\,v_{\text{rel}}} = 1.489\, t_{\text{Pl}}. \tag{25}$$

The second Heisenberg bound:

$$E_{\text{cell}}^{\text{lower}} = g_* \cdot \frac{\hbar}{2\tau_{gg}} = 35.86\, E_{\text{Pl}}. \tag{26}$$

Cauchy uncertainty propagation from $\delta\alpha_s = 0.0010$ (PDG 2024): $\delta E/E = 2\delta\alpha_s/\alpha_s^{\text{MZ}} \times (\alpha_s/\alpha_s^{\text{MZ}}) \approx 1.70\%$, giving $\delta E_{\text{cell}} = \pm 0.61\, E_{\text{Pl}}$.

12.3 Statistical significance

Table 2 places E_{cell} in the hierarchy of known energy scales. The result is derived from two axioms with $g_* = 106.75$ as the only experimental input.

Table 2: Energy scale comparisons. $E_{\text{cell}} = 1.044 \times 10^{11}$ J is derived from Axioms 1 and 2 with $g_* = 106.75$ as the only experimental input.

Reference scale	Energy (J)	E/E_{cell}
Electron $m_e c^2$	8.19×10^{-14}	7.8×10^{-25}
Proton $m_p c^2$	1.50×10^{-10}	1.4×10^{-21}
LHC full beam	3.15×10^{-3}	3.0×10^{-14}
Planck energy E_{Pl}	1.956×10^{9}	1/53.4
E_{cell} **(TUO, two axioms)**	$\mathbf{1.044 \times 10^{11}}$	**1**
Solar mass-energy $M_\odot c^2$	1.79×10^{47}	1.7×10^{36}

13 Pre-Emergence Temperature and the $15/\pi^2$ Factor

Theorem 13.1 (T_{TUO} and $15/\pi^2$)**.** *The pre-emergence temperature is*

$$T_{\text{TUO}} = \left(\frac{15}{\pi^2}\right)^{1/4} T_{\text{Pl}} \approx 1.108\, T_{\text{Pl}}, \tag{27}$$

independent of g_*. Moreover, $T_{\text{TUO}} = T_{\text{upper}}$: the upper energy bound and the thermal equilibrium temperature coincide exactly.

Proof. Set $\rho_{\text{HB}} = g_* E_{\text{Pl}}/(2\ell_{\text{Pl}}^3)$ equal to $\rho_{\text{SB}}(T_{\text{TUO}}) = (\pi^2/30) g_* (k_{\text{B}} T_{\text{TUO}})^4/(\hbar c)^3$. g_* cancels on both sides (explaining g_*-independence). Using $E_{\text{Pl}} = \hbar c/\ell_{\text{Pl}}$: $(k_{\text{B}} T_{\text{TUO}})^4 = (15/\pi^2)(k_{\text{B}} T_{\text{Pl}})^4$. Direct numerical evaluation: $T_{\text{upper}}/T_{\text{Pl}} = 1.110319$, $T_{\text{TUO}}/T_{\text{Pl}} = 1.110319$ (relative difference $< 10^{-15}$). □

Remark 13.2 (Correct interpretation of $15/\pi^2$). The factor $15/\pi^2$ is the *energy-density ratio* $\rho_{\text{HB}}/\rho_{\text{SB}}(T_{\text{Pl}})$. It is *not* the particle-number non-equilibrium ratio, which equals $N_{\text{HB}}/N_{\text{th}} \approx 6.7$–$9.0$ (Section 14). These are distinct physical quantities. The g_*-independence is non-trivial: both energies scale identically with g_*, so the ratio and hence the temperature are universal.

14 QGP at Emergence: Free-Streaming Regime

Theorem 14.1 (QGP Free-Streaming)**.** *At emergence, the QGP is in the* free-streaming *regime with mean free path $\lambda_{\text{mfp}} \approx 85$–$94\,\ell_{\text{Pl}} \gg \ell_{\text{Pl}}$. Thermalisation occurs only at $\tau_{\text{th}} \approx 85$–$94\,t_{\text{Pl}}$.*

The QGP temperature range follows from equating the Stefan-Boltzmann energy density to the Planck-cell energy density at each bound:

$$T \in [1.005,\ 1.108]\, T_{\text{Pl}}. \tag{28}$$

The leading-order perturbative QCD mean free path:

$$\lambda_{\rm mfp} = \frac{\hbar c}{\alpha_s\, k_{\rm B} T} = 85\text{–}94\,\ell_{\rm Pl}. \tag{29}$$

The Debye screening length:

$$\lambda_D = \frac{\hbar c}{\sqrt{g^2 T^2 (N_c/3 + N_f/6)}} = 1.75\text{–}1.93\,\ell_{\rm Pl}, \tag{30}$$

confirming colour screening is sub-Planck-scale while particle transport is free-streaming.

Three timescales characterise the emergence event:

$$\tau_{\rm int} = 1.489\,t_{\rm Pl} \qquad \text{(first QCD interaction)}, \tag{31}$$

$$t_{\rm class} = \sqrt{300}\,t_{\rm Pl} \approx 17.3\,t_{\rm Pl} \qquad (\delta V/V < 1\%\text{, classical description valid}), \tag{32}$$

$$\tau_{\rm th} \approx 85\text{–}94\,t_{\rm Pl} \qquad \text{(full thermalisation)}. \tag{33}$$

Remark 14.2. The initial state is highly non-equilibrium: particle number ratio $N_{\rm HB}/N_{\rm th}(T_{\rm TUO}) \approx$ 6.7–9.0. Boltzmann kinetic equations govern dynamics in $[0, \tau_{\rm th}]$. This corrects an implicit earlier assumption of instantaneous thermalisation.

15 Flat-Space Equation of Motion: $\ddot{\sigma} = c^2/\sigma$

Theorem 15.1 (Flat-Space EOM)**.** *In flat Minkowski spacetime, the equation of motion for the QGP radiation shell is*

$$\boxed{\ddot{\sigma} = \frac{c^2}{\sigma}}, \tag{34}$$

universal and independent of E_0, g_*, and $\ell_{\rm Pl}$.

Proof. We use Newton's second law with SR inertia — no GR.

Step 1. Flat-space Doppler redshift: $E_{\rm rad}(\sigma) = E_0(\ell_{\rm Pl}/\sigma)$.

Step 2. Energy density: $\rho(\sigma) = 3E_0\ell_{\rm Pl}/(4\pi\sigma^4)$.

Step 3. QGP pressure and force: $P = \rho/3$, $F = P \cdot 4\pi\sigma^2 = E_0\ell_{\rm Pl}/\sigma^2$.

Step 4. SR inertial mass: $m_{\rm eff} = E_{\rm rad}/c^2 = E_0\ell_{\rm Pl}/(\sigma c^2)$.

Step 5. $m_{\rm eff}\ddot{\sigma} = F$: $\dfrac{E_0\ell_{\rm Pl}}{\sigma c^2}\ddot{\sigma} = \dfrac{E_0\ell_{\rm Pl}}{\sigma^2}$. E_0 and $\ell_{\rm Pl}$ cancel completely, giving (34). □

Analytical solution (energy integral with $\sigma(0) = \ell_{\rm Pl}$, $\dot{\sigma}(0) = 0$):

$$\dot{\sigma}^2 = 2c^2 \ln\left(\frac{\sigma}{\ell_{\rm Pl}}\right). \tag{35}$$

15.1 Why positive pressure accelerates here but decelerates in GR

- **GR (Friedmann):** P enters the source term $\rho + 3P$ of the Raychaudhuri equation. For $w = 1/3$: $\ddot{a}/a = -(4\pi G/3)(\rho + 3P) = -(4\pi G/3) \cdot 2\rho < 0$. *Gravitational deceleration.*

- **Flat Minkowski (TUO):** P acts as a mechanical force $F = P \times A$ outward on the shell boundary. Positive P drives outward acceleration. *Mechanical acceleration.*

These are different physical mechanisms in different geometric settings.

16 Deceleration Parameter and De Sitter Equivalence

Theorem 16.1 ($q(t) = -(\ell_{\rm Pl}/ct)^2$)**.** *The deceleration parameter of the TUO wavepacket is*

$$q(t) = -\left(\frac{\ell_{\rm Pl}}{ct}\right)^2 < 0 \quad \forall\, t > 0, \tag{36}$$

with $q(t_{\rm Pl}) = -1$ exactly (de Sitter equivalent).

Proof. The wavepacket from Heisenberg position bound with initial speed c: $\sigma(t) = \ell_{\rm Pl}\sqrt{1 + (ct/\ell_{\rm Pl})^2}$. Differentiating: $\dot{\sigma} = c(ct/\ell_{\rm Pl})/\sqrt{1 + (ct/\ell_{\rm Pl})^2}$, $\ddot{\sigma} = c^2\ell_{\rm Pl}^2/(\ell_{\rm Pl}^2 + c^2t^2)^{3/2}$. Then:

$$q = -\frac{\ddot{\sigma}\,\sigma}{\dot{\sigma}^2} = -\frac{c^2\ell_{\rm Pl}^3/(\ell_{\rm Pl}^2 + c^2t^2)}{c^4t^2/(\ell_{\rm Pl}^2 + c^2t^2)} = -\frac{\ell_{\rm Pl}^2}{c^2t^2}. \tag{37}$$

At $t = t_{\rm Pl}$: $ct_{\rm Pl} = \ell_{\rm Pl}$ (exact, by definition of Planck units), so $q(t_{\rm Pl}) = -1$. □

The wavepacket velocity $v(t) = c(ct/\ell_{\rm Pl})/\sqrt{1 + (ct/\ell_{\rm Pl})^2} < c$ for all t; at $t = t_{\rm Pl}$: $v(t_{\rm Pl}) = c/\sqrt{2} \approx 0.707\,c$. Standard inflation requires super-luminal expansion of the scale factor; TUO expansion is sub-luminal at all times.

17 Expansion Law and Equation of State

Theorem 17.1 (Expansion law)**.**

$$\sigma(t) = \ell_{\rm Pl}\sqrt{1 + \left(\frac{ct}{\ell_{\rm Pl}}\right)^2}, \qquad V(t) = \frac{4\pi}{3}\,\ell_{\rm Pl}^3\left[1 + \left(\frac{ct}{\ell_{\rm Pl}}\right)^2\right]^{3/2}. \tag{38}$$

Quantum correction for $ct \gg \ell_{\rm Pl}$: $\Delta V/V = 3(\ell_{\rm Pl}/ct)^2$.

Theorem 17.2 ($w = 1/3$ derived)**.** *The equation of state $w = 1/3$ is derived, not assumed.*

Proof. $\Delta p = \hbar/(2\ell_{\mathrm{Pl}}) = M_{\mathrm{Pl}}c/2$. Relativistic velocity: $v_0 = pc^2/E = (M_{\mathrm{Pl}}c/2)c^2/(E_{\mathrm{Pl}}/2) = c$. Since $v_0 = c$, all emergent particles are ultra-relativistic with $w = P_{\mathrm{kinetic}}/(\rho c^2) = 1/3$. □

17.1 Hubble parameter

$$H_{\mathrm{TUO}}(t) = \frac{c^2 t}{\ell_{\mathrm{Pl}}^2 + c^2 t^2}. \tag{39}$$

At $t = t_{\mathrm{Pl}}$ using $\ell_{\mathrm{Pl}} = ct_{\mathrm{Pl}}$:

$$H_{\mathrm{TUO}}(t_{\mathrm{Pl}}) = \frac{c^2 t_{\mathrm{Pl}}}{2c^2 t_{\mathrm{Pl}}^2} = \frac{1}{2t_{\mathrm{Pl}}} = H_{\mathrm{FRW}}(t_{\mathrm{Pl}}). \tag{40}$$

18 Junction Conditions

Theorem 18.1 (TUO–FRW Junction)**.** *H, w, and k are all continuous at the junction. The junction time is*

$$t_{\mathrm{junc}} \in [17.3,\ 152]\, t_{\mathrm{Pl}}, \tag{41}$$

not at t_{Pl} as stated in earlier presentations. The lower bound ($17.3\, t_{\mathrm{Pl}}$) is set by $\delta V/V < 1\%$; the upper bound ($152\, t_{\mathrm{Pl}}$) is the thermalisation time with realistic FRW cooling $T(t) \propto t^{-1/2}$ (isothermal estimate: 85–94 t_{Pl}).

Proof. (i) H continuous: Eq. (40). ✓(ii) $w = 1/3$ on both sides: Theorem 17.2. ✓(iii) $k = 0$: Axiom 1. ✓

Classical limit. The quantum volume uncertainty $\delta V/V = 3(\ell_{\mathrm{Pl}}/ct)^2$ drops below 1% when $t > \sqrt{300}\, t_{\mathrm{Pl}} \approx 17.3\, t_{\mathrm{Pl}}$.

Thermalisation. Full QGP thermalisation requires $\tau_{\mathrm{th}} = \lambda_{\mathrm{mfp}}/c \approx 85$–$94\, t_{\mathrm{Pl}}$ (Theorem 14.1). Both bounds together give $t_{\mathrm{junc}} \in [17, 94]\, t_{\mathrm{Pl}}$. □

Remark 18.2. The energy density at the junction differs from the FRW thermal value by the factor $15/\pi^2 \approx 1.52$ (Section 13). This does not break continuity of H, w, or k; it reflects the difference between the TUO Heisenberg-minimum state and the FRW thermal-equilibrium state. As the initial maximum-fluctuation state thermalises, the effective temperature relaxes to the value consistent with the Friedmann equation.

19 Simultaneous Emergence: A Possible Mechanism for CMB Uniformity

Remark 19.1. The analysis in this section is a *possible mechanism*, not a proven theorem. The saddle-point argument below suggests that simultaneous emergence of all Planck cells is exponentially preferred, offering a possible explanation for the observed CMB uniformity, but a complete proof with controlled error bounds is left as Open Question 23.2.

Partition the Minkowski background into a lattice of Planck cells indexed $i = 1, \ldots, N$ where $N = (R_0\sqrt{t_{\rm Pl}/t_0}/\ell_{\rm Pl})^3 \approx 8.8 \times 10^{92}$. The constrained Euclidean path integral is

$$Z = \int \mathcal{D}\phi \; e^{-S_E[\phi]} \prod_k \delta\big(Q_k[\phi]\big). \tag{42}$$

Writing $\delta(Q_k) = \int \frac{d\lambda_k}{2\pi} e^{i\lambda_k Q_k}$ and evaluating at the saddle point: single-cell fluctuations against $N-1$ vacuum cells incur action penalty $\Delta S \sim N$, while simultaneous N-cell emergence satisfies the global constraint with $\Delta S = 0$. The ratio of contributions is formally

$$\frac{Z_{\rm simultaneous}}{Z_{\rm single}} \sim e^N = e^{8.8\times 10^{92}}. \tag{43}$$

This suggests simultaneous emergence is the overwhelmingly dominant configuration.

20 Exotic Phenomena Permitted by the Zero-Sum Axiom

Table 3: Phenomena permitted by Axiom 2. ✓ = observed; ∼ = conjectural direction consistent with ZSC.

Phenomenon	Zero-sum mechanism	Status
Virtual particles	Pair: $Q_{\rm pair} = 0$; Heisenberg limits duration, not existence.	✓
Entanglement	Singlet enforces $\langle S_z\rangle_A + \langle S_z\rangle_B = 0$.	✓
CPT symmetry	Every process creating particles has CPT-conjugate; $B - L = 0$ maintained.	✓
BH information	$S_{\rm matter} + S_{\rm grav} = \text{const}$: information transfers to gravitational sector.	∼
Dark sector	Any hidden sector with internal $Q_{\rm dark} = 0$, $B_{\rm dark} - L_{\rm dark} = 0$ satisfies ZSC.	∼
Parallel configs	Other $N_{\rm gen}$ rows of Table 1 satisfy ZSC.	∼

21 Hot Big Bang Initial Conditions

Theorem 21.1 (TUO-to-HBB handoff)**.** *The maximum-fluctuation emergence delivers all five HBB initial conditions without additional assumptions.*

Proof. (i) $w = 1/3$: Theorem 17.2. ✓(ii) $H(t_{\rm Pl}) = 1/(2t_{\rm Pl})$: Eq. (40). ✓(iii) $\Omega = 1$: Axiom 1. ✓(iv) Full SM plasma: $E_{\rm cell} = 1.044 \times 10^{11}$ J $\gg 2m_t c^2 = 5.46 \times 10^{-8}$ J. Ratio $\approx 1.9 \times 10^{18}$; all species produced. ✓(v) Temperature: $T_{\rm TUO} = (15/\pi^2)^{1/4} T_{\rm Pl} \approx 1.108\, T_{\rm Pl}$ (Theorem 13.1). ✓ □

TUO ($t < t_{\rm Pl}$)	**at** $t_{\rm junc}$	**Hot Big Bang** ($t > t_{\rm junc}$)
Zero-sum pre-emergence	$\longleftrightarrow$	Radiation domination
Wavepacket $\sigma(t)$, $v < c$		$a(t) \propto t^{1/2}$
$w = 1/3$ (derived)	$w = 1/3$	$w = 1/3$ (assumed)
$H = 1/(2t_{\rm Pl})$ (derived)	H continuous	$H = 1/(2t)$
$k = 0$ (Axiom I)	$\Omega = 1$	$\Omega = 1$ (assumed)
$g_* = 106.75$ (all SM)	plasma	HBB thermodynamics

22 Quantitative Predictions

23 Open Problems

Open Question 23.1 (Baryon asymmetry)*.* TUO shows that $B - L = 0$ permits matter-only configurations. It does not derive $\eta = 6.12 \times 10^{-10}$. A full derivation requires QFT baryon-number dynamics at $T_{\rm TUO}$ with Sakharov-type washout factors.

Open Question 23.2 (CMB power spectrum)*.* The saddle-point argument of Section 19 suggests simultaneous emergence, giving $\Delta T/T = 0$ at leading order. Within-cell quantum fluctuations give $\Delta E_{\rm cell}/E_{\rm cell} \sim 1/\sqrt{g_*} \approx 9.7\%$. The observed $\Delta T/T \sim 10^{-5}$ requires ~ 4 orders of magnitude of suppression through GR dynamics. The specific calculation required is the two-point stress-energy correlator $\langle T_{\mu\nu}(x) T_{\rho\sigma}(x')\rangle_{\rm ZS}$ in a Planck-cell volume, propagated to horizon crossing.

Open Question 23.3 (Residual dark energy)*.* Zero-sum predicts $\Lambda_{\rm eff} = 0$ at leading order. The residual $\Lambda_{\rm obs} \approx 10^{-122} E_{\rm Pl}^4$ is not derived here and requires the companion QGD framework.

Open Question 23.4 (Precise junction time)*.* The range $t_{\rm junc} \in [17, 94]\, t_{\rm Pl}$ is established; the precise value requires full Boltzmann-equation treatment of the QGP.

Open Question 23.5 (SM particle content from axioms)*.* TUO uses $g_* = 106.75$ from experiment. Whether g_* and $N_{\rm gen} = 3$ follow from the constrained path integral is open.

Table 4: TUO results from Axioms 1–2 and SM content. ✓✓ = TUO derives what standard cosmology assumes; ✓ = quantitatively consistent; ∼ = qualitatively consistent; ⋆ = new distinguishing prediction; † = open problem.

#	Result	TUO value	Observation	Status
1	E_{tot}	0 exact (Thm 10.1)	≈ 0 (flat)	✓✓
2	Ω	1 exact (Axiom 1)	1.0007 ± 0.0037 [1]	✓✓
3	w	1/3 derived (Thm 17.2)	1/3 (FRW input)	✓✓
4	$H(t_{\text{Pl}})$	$1/(2t_{\text{Pl}})$ derived	$1/(2t_{\text{Pl}})$ (FRW input)	✓✓
5	Expansion law	$V \propto t^3$, $v < c$	$V \propto t^3$ (rad. dom.)	✓✓ + ⋆
6	E_{cell}	$(g_*/2)E_{\text{Pl}} = 1.04 \times 10^{11}$ J	6.87×10^{10} J (thermal)	✓
7	λ_{mfp}	85–94 ℓ_{Pl} (Thm 14.1)	not yet measured	⋆
8	$q(t_{\text{Pl}})$	-1 exact (Thm 16.1)	de Sitter	⋆
9	t_{junc}	$[17.3, 152]\, t_{\text{Pl}}$	not yet measured	⋆
10	$B - L = 0$ per gen	SM structure (Thm 8.1)	anomaly cancellation	✓
11	Baryon asymmetry	$B = L \neq 0$ permitted	$\eta = 6.12 \times 10^{-10}$	∼ †
12	CMB uniformity	simultaneous emergence	$\Delta T/T \sim 10^{-5}$	∼ †

Open Question 23.6 (SM gauge group from axioms). SU(3) × SU(2) × U(1) is input, not derived.

Retraction. The formula $x_\star = \sqrt{g_*/2}$ (originally proposed for the spectral index) was found to have a coefficient error and is withdrawn. The spectral index n_s is not derived in TUO.

24 Summary

TUO is a pre-Big-Bang framework based on two axioms applied to the infinite-dimensional Fock space of Standard Model fields. Its central object is the infinite zero matrix $\mathbf{0}_\infty$; its central equation is $\mathbf{Q}[\hat{\rho}] = \mathbf{0}_\infty$.

Derived results (from axioms, $g_* = 106.75$, PDG 2024).

1. $w = 1/3$ — radiation equation of state
2. $H(t_{\text{Pl}}) = 1/(2t_{\text{Pl}})$ — Planck-era Hubble rate
3. $\Omega = 1$ — exact spatial flatness

4. $V(t) \propto t^3$, $v < c$ always — expansion law
5. $E_{\rm cell} = (g_*/2)E_{\rm Pl}$; $E_{\rm total} = 0$ — zero energy
6. No-annihilation theorem — matter-only stability
7. $B - L = 0$, $Q = 0$ per SM generation — anomaly identity
8. $T_{\rm TUO} = (15/\pi^2)^{1/4} T_{\rm Pl} \approx 1.108\, T_{\rm Pl}$ — universal temperature
9. $E_{\rm cell} \in [35.86 \pm 0.61,\ 53.375]\, E_{\rm Pl}$ — two Heisenberg bounds
10. $\lambda_{\rm mfp} = 85\text{–}94\, \ell_{\rm Pl}$ — free-streaming QGP
11. $\ddot{\sigma} = c^2/\sigma$ in flat Minkowski (Newton+SR)
12. $q(t) = -(\ell_{\rm Pl}/ct)^2$; $q(t_{\rm Pl}) = -1$ exact — de Sitter equivalent
13. $t_{\rm junc} \in [17, 94]\, t_{\rm Pl}$ — corrected junction timing
14. All 5 HBB initial conditions — seamless handoff

Open problems. Baryon asymmetry η; CMB power spectrum amplitude; residual dark energy $\Lambda_{\rm obs}$; precise junction time; SM particle content from axioms; $N_{\rm gen} = 3$ derivation.

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A Planck Units and Key Identities

$$
\begin{aligned}
E_{\rm Pl} &= \sqrt{\hbar c^5/G} = 1.956 \times 10^9 \text{ J}, & \ell_{\rm Pl} &= \sqrt{\hbar G/c^3} = 1.616 \times 10^{-35} \text{ m}, \\
t_{\rm Pl} &= \sqrt{\hbar G/c^5} = 5.391 \times 10^{-44} \text{ s}, & T_{\rm Pl} &= \sqrt{\hbar c^5/(G k_{\rm B}^2)} = 1.417 \times 10^{32} \text{ K}, \\
M_{\rm Pl} &= \sqrt{\hbar c/G} = 2.177 \times 10^{-8} \text{ kg}.
\end{aligned} \tag{44}
$$

Two exact identities used throughout (verified to 15 significant figures in `tuo_complete_theory.py`):

$$
c\, t_{\rm Pl} = \ell_{\rm Pl}, \qquad G M_{\rm Pl}^2 = \hbar c. \tag{45}
$$

B Running Couplings at the Planck Scale

QCD (1-loop):

$$\alpha_s(\mu^2) = \frac{\alpha_s(M_Z^2)}{1 + \frac{b_0}{2\pi}\alpha_s(M_Z^2)\ln(\mu^2/M_Z^2)}, \quad b_0 = 7\ (n_f = 6). \tag{46}$$

With $\alpha_s(M_Z) = 0.1179$, $M_Z = 91.1876\,\text{GeV}$: $\alpha_s(E_{\text{Pl}}/2) = 0.01055 \pm 0.0002$.

QED (1-loop):

$$\alpha(\mu^2) = \frac{\alpha_0}{1 - \frac{\alpha_0}{3\pi}\ln(\mu^2/m_e^2)}, \quad \alpha_0 = 1/137.036. \tag{47}$$

At $\mu = E_{\text{Pl}}/2$: $\alpha = 0.00792$.

C Derivation of $q(t) = -(\ell_{\text{Pl}}/ct)^2$

With $x \equiv ct/\ell_{\text{Pl}}$:

$$\dot{\sigma} = \frac{cx}{\sqrt{1+x^2}}, \qquad \ddot{\sigma} = \frac{c^2}{\ell_{\text{Pl}}(1+x^2)^{3/2}}. \tag{48}$$

$$q = -\frac{\ddot{\sigma}\,\sigma}{\dot{\sigma}^2} = -\frac{c^2/(1+x^2)^{3/2}\cdot\sqrt{1+x^2}}{c^2x^2/(1+x^2)} = -\frac{1}{x^2} = -\frac{\ell_{\text{Pl}}^2}{c^2t^2}. \tag{49}$$

At $t = t_{\text{Pl}}$: $x = ct_{\text{Pl}}/\ell_{\text{Pl}} = 1$, so $q = -1$.

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